

Akhil Prasad Chinthala

AI / ML Engineer | LLM & Computer Vision Systems

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SUMMARY

Final-year IT student (CGPA 8.64) shipping LLM-powered features as an AI Engineering Intern at AMIK Technologies and authoring agentic coding benchmarks for frontier-model evaluation as a freelance AI trainer at Handshake AI. Hands-on across Python backend and applied AI: FastAPI model-serving endpoints, retrieval-augmented generation and LLM evaluation, deep-learning computer vision at 90%+ accuracy on FER-2013, and real-time inference under 50 ms. Seeking AI/ML or Python backend engineering roles building production LLM and vision systems.

EXPERIENCE

AI Engineering Intern

Apr 2026 – Present

AMIK Technologies

Hyderabad, India

- Build production retrieval-augmented generation (RAG) pipelines over internal documents — chunking, embedding generation, and vector similarity search — replacing manual lookup with grounded, citation-backed answers.
- Design and version prompt templates with structured outputs and function calling, standardising LLM responses into schema-validated JSON consumed directly by downstream services.
- Ship an automated evaluation harness scoring retrieval relevance, answer faithfulness, and latency across model and prompt versions — turning subjective review into regression-tested releases.
- Serve models behind FastAPI endpoints with token streaming, per-request cost logging, and provider fallback for resilience under rate limits.
- Reduce inference cost and latency through chunk-size tuning, semantic caching, and context compression; integrate features into the live product with the backend team.

Freelance AI Trainer — Agentic Benchmark Authoring

Jul 2026 – Present

Handshake AI

Remote

- Author terminal-based benchmark tasks used to evaluate frontier AI coding agents — 21 tasks delivered across 7 domains, including systems infrastructure, model-training and ML pipelines, data processing/ETL, databases, and interactive simulation.
- Ship each task as a reproducible package — natural-language specification, Dockerised environment, automated test-based verifier, and a working reference solution — built to clear automated QA gates and human review.

Software Development Intern — Python

2024

Oasis Infobyte

Remote

- Delivered five Python applications on a structured internship track, each specified, built, and demoed independently.
- Highlights: a speech-recognition voice assistant that parses spoken intent into system commands, a REST-integrated weather CLI, and a multi-client TCP chat server handling concurrent sessions.

PROJECTS

CAG Real-Time Emotion Tracker | PyTorch, InceptionV3, OpenCV, Python

2024 – 2025

- Fine-tuned an attention-guided InceptionV3 model on FER-2013 to **90%+ accuracy across 7 emotion classes**, well ahead of the baseline CNN.
- Engineered a Cache-Augmented Generation pipeline — batch inference plus LRU caching of repeated frames — holding **end-to-end latency under 50 ms** on live webcam input.
- Led a 3-person team from architecture through delivery; presented results to a faculty review panel.

Full-Stack Booking Platform with Voice AI Assistant | React, Vite, Node.js, Express, Web Speech API

2025

- Built an end-to-end web application — React/Vite frontend backed by an Express REST API — deployed live in production for a real business client (example: a dental clinic), handling genuine customer bookings.
- Integrated a voice-enabled AI assistant (*DentalBot*) that answers the client's service FAQs and converts visitors into bookings.
- Automated the full booking workflow: appointment data through the REST API, Google Sheets sync for the client's staff, and automated email confirmations.

Hand-Gesture Media Controller | MediaPipe, OpenCV, Python, AppleScript

2024

- Mapped MediaPipe hand-landmark detection to macOS volume and playback controls for fully touchless media operation.
- Sustained **30 FPS with sub-20 ms gesture-response latency** by optimising the landmark-to-action inference loop.

- Responsive grocery storefront in vanilla JavaScript with dynamic filtering, cart state management, and a full checkout flow.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Java, C, C++

LLM & Generative AI: Retrieval-Augmented Generation (RAG), prompt engineering, embeddings & vector search, function calling / structured outputs, LLM evaluation, semantic caching

Machine Learning & CV: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, OpenCV, MediaPipe, NumPy, Pandas, fine-tuning

Deployment & Tooling: FastAPI, REST APIs, Git & GitHub, Linux (Ubuntu), model serving, latency & cost profiling

Web & Data: React, Vite, Node.js, Express, MySQL, MongoDB, vector databases

Foundations: Deep Learning, Computer Vision, NLP, Data Structures & Algorithms, Operating Systems, Networks

Spoken languages: Telugu (Native), English (Fluent), Hindi (Conversational)

EDUCATION

Matrusri Engineering College — B.Tech, Information Technology — CGPA 8.64

2023 – 2027

Sri Chaitanya Junior Kalasala — Higher Secondary, Science & Mathematics — 93%

2023

The Hyderabad Public School — Secondary School — 89%

2021